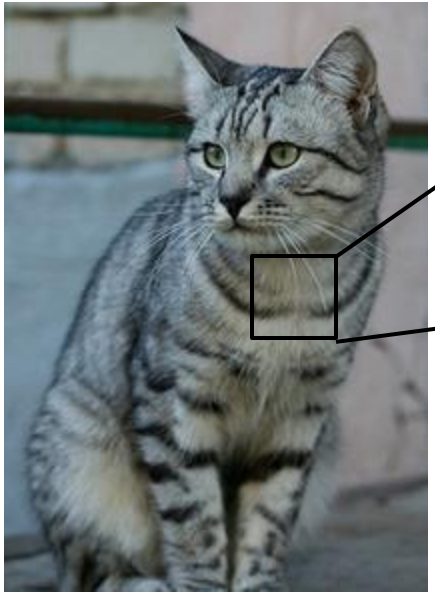
Image Classification with Linear Classifiers

Image Classification: A core task in Computer Vision

(assume given a set of possible labels)
{dog, cat, truck, plane, ...}

→ cat

The Problem: Semantic Gap



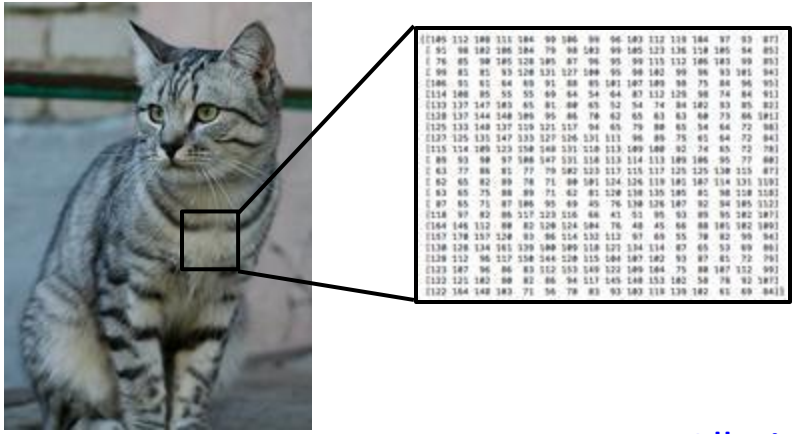
```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 110 105  94  85]
 [ 76  85  90 105 128 105  87  96  95  99 115 112 106 103  99  85]
 [ 99  81  81  93 120 131 127 100  95  98 102  99  96  93 101  94]
 [106  91  61  64  69  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  69  64  54  64  87 112 129  98  74  84  91]
 [133 137 147 103  65  81  80  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  62  65  63  63  60  73  86 101]
 [125 133 148 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 109 123 150 148 131 118 113 109 100  92  74  65  72  78]
 [ 89  93  90  97 100 147 131 118 113 114 113 109 106  95  77  80]
 [ 63  77  86  81  77  79 102 123 117 115 117 125 125 130 115  87]
 [ 62  65  82  89  78  71  80 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
 [ 87  65  71  87 106  95  69  45  76 130 126 107  92  94 105 112]
 [118  97  82  86 117 123 116  66  41  51  95  93  89  95 102 107]
 [164 146 112  80  82 120 124 104  76  48  45  66  88 101 102 109]
 [157 170 157 120  93  86 114 132 112  97  69  55  70  82  99  94]
 [130 128 134 161 139 100 109 118 121 134 114  87  65  53  69  86]
 [128 112  96 117 150 144 120 115 104 107 102  93  87  81  72  79]
 [123 107  96  86  83 112 153 149 122 109 104  75  80 107 112  99]
 [122 121 102  80  82  86  94 117 145 148 153 102  58  78  92 107]
 [122 164 148 103  71  56  78  83  93 103 119 139 102  61  69  84]]
```

What the computer sees

An image is a tensor of integers between [0, 255]:

e.g. 800 x 600 x 3
(3 channels RGB)

Challenges: Viewpoint variation



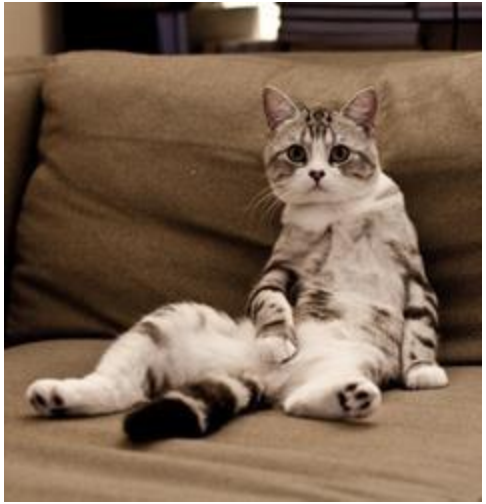
All pixels change when the camera moves!

Challenges: Illumination

Challenges: Background Clutter

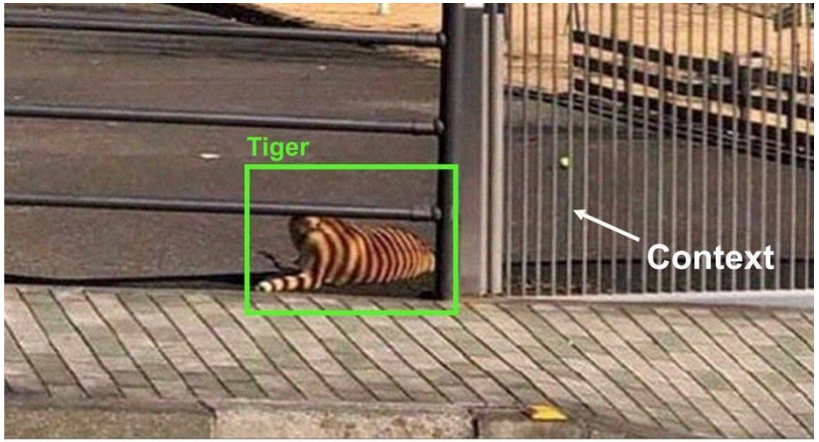
Challenges: Occlusion

Challenges: Deformation



Challenges: Intraclass variation

Challenges: Context



An image classifier

```
def classify_image(image):
  # Some magic here?
  return class_label
```

Unlike e.g. sorting a list of numbers,

no obvious way to hard-code the algorithm for recognizing a cat, or other classes.

First classifier: Nearest Neighbor

```
def train(images, labels):
  # Machine learning!
  return model
```

→ Memorize all data and labels

```
def predict(model, test_images):
  # Use model to predict labels
  return test_labels
```

→ Predict the label of the most similar training image

First classifier: Nearest Neighbor

deer bird plane cat car

Training data with labels

query data

Distance Metric $\left| \text{ }, \text{ } \right| \rightarrow \mathbb{R}$

Distance Metric to compare images

L1 distance: $$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

test image

56	32	10	18
90	23	128	133
24	26	178	200
2	0	255	220

-

training image

10	20	24	17
8	10	89	100
12	16	178	170
4	32	233	112

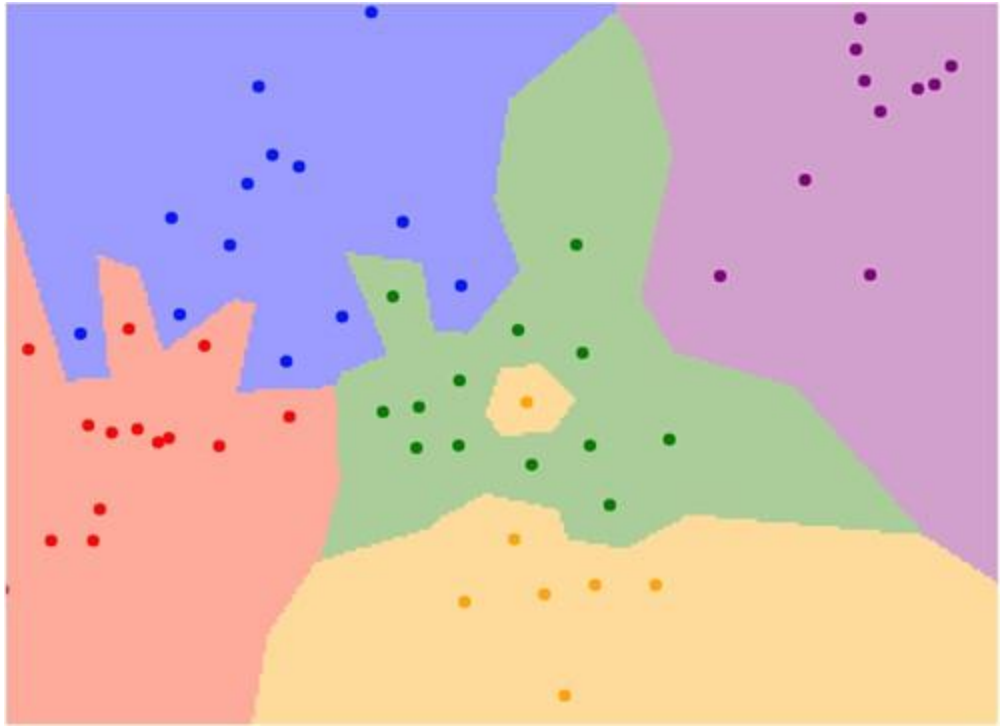
=

pixel-wise absolute value differences

46	12	14	1
82	13	39	33
12	10	0	30
2	32	22	108

$\xrightarrow{\text{add}}$ 456

What does this look like?



1-nearest neighbor

K-Nearest Neighbors

Instead of copying label from nearest neighbor,
take majority vote from K closest points

K = 1

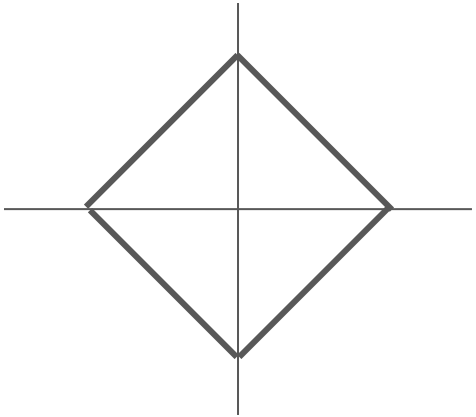
K = 3

K = 5

K-Nearest Neighbors: Distance Metric

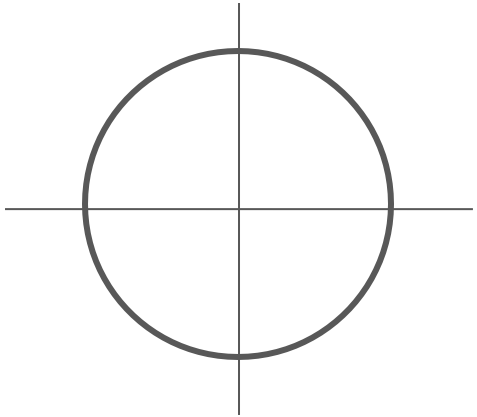
L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p \left| I_1^p - I_2^p \right|$$



L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p \left(I_1^p - I_2^p \right)^2}$$



K-Nearest Neighbors: Distance Metric

L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

K = 1

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p \left(I_1^p - I_2^p\right)^2}$$

K = 1

Hyperparameters

- What is the best value of k to use? What is the best distance to use?

- These are hyperparameters: choices about the algorithms themselves.

- Very problem/dataset-dependent.
- Must try them all out and see what works best.

Setting Hyperparameters

Idea #1: Choose hyperparameters that work best on the training data

train

Setting Hyperparameters

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BAD: K = 1 always works perfectly on training data

train

Setting Hyperparameters

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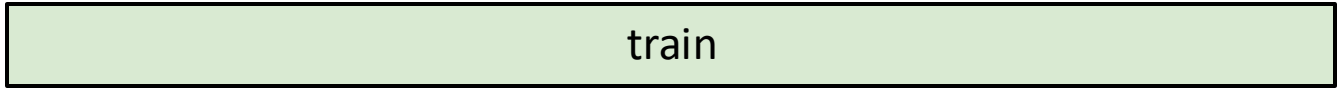
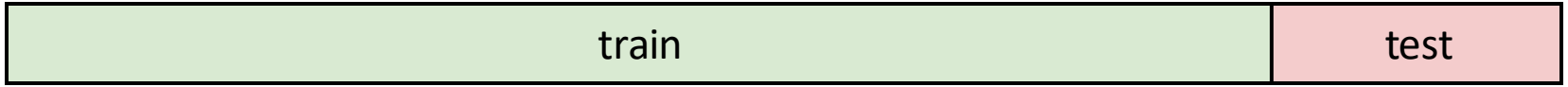


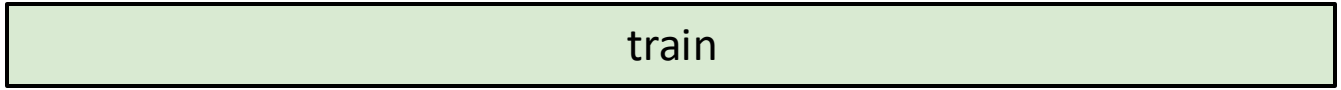
Idea #2: choose hyperparameters that work best on test data



Setting Hyperparameters

Idea #1: Choose hyperparameters that work best on the training data

BAD: K = 1 always works perfectly on training data



Idea #2: choose hyperparameters that work best on test data

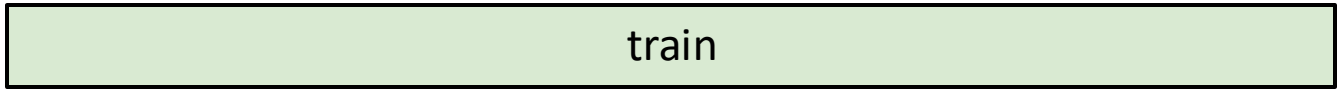
BAD: No idea how algorithm will perform on new data

Never do this!

Setting Hyperparameters

Idea #1: Choose hyperparameters that work best on the training data

BAD: K = 1 always works perfectly on training data



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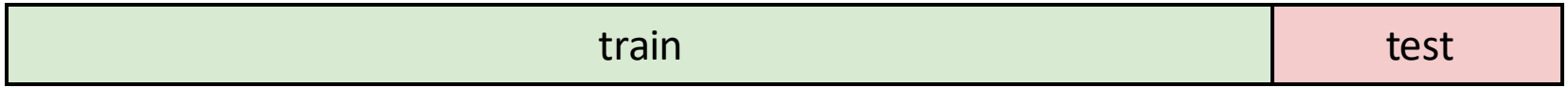

Idea #3: Split data into train, val; choose hyperparameters on val and evaluate on test

Better!

Setting Hyperparameters

train

Idea #4: Cross-Validation: Split data into folds, try each fold as validation and average the results

fold 1	fold 2	fold 3	fold 4	fold 5
fold 1	fold 2	fold 3	fold 4	fold 5
fold 1	fold 2	fold 3	fold 4	fold 5
fold 1	fold 2	fold 3	fold 4	fold 5
fold 1	fold 2	fold 3	fold 4	fold 5

test

Useful for small datasets, but not used too frequently in deep learning

Example Dataset: CIFAR10

10 classes
50,000 training images
10,000 testing images

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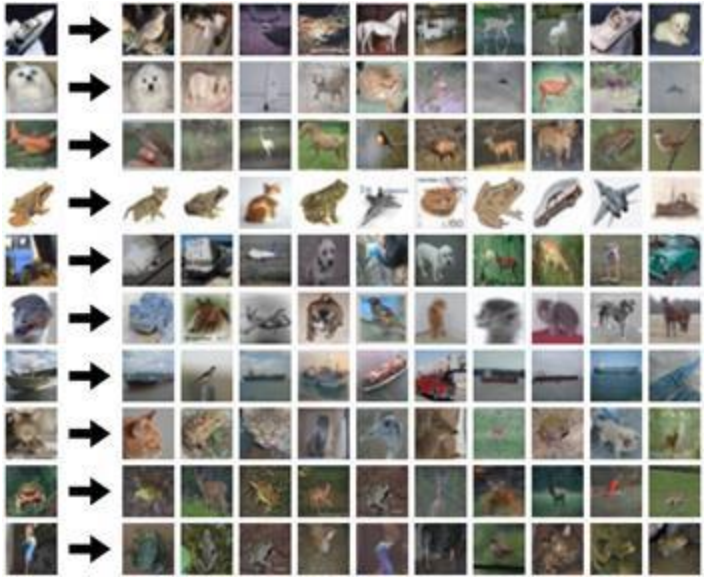
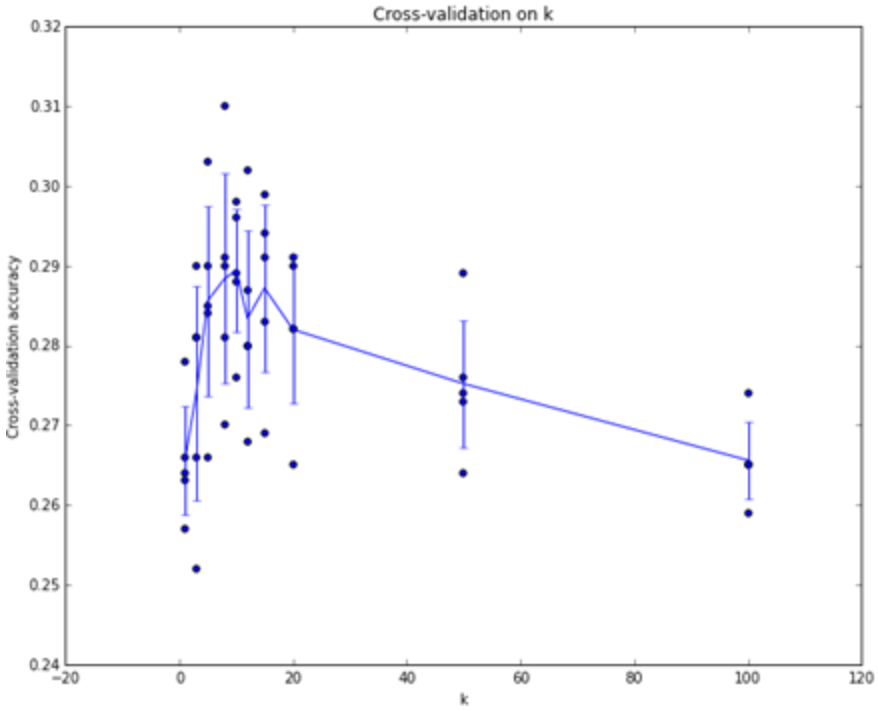
10 classes
50,000 training images
10,000 testing images

Test images and nearest neighbors



Setting Hyperparameters



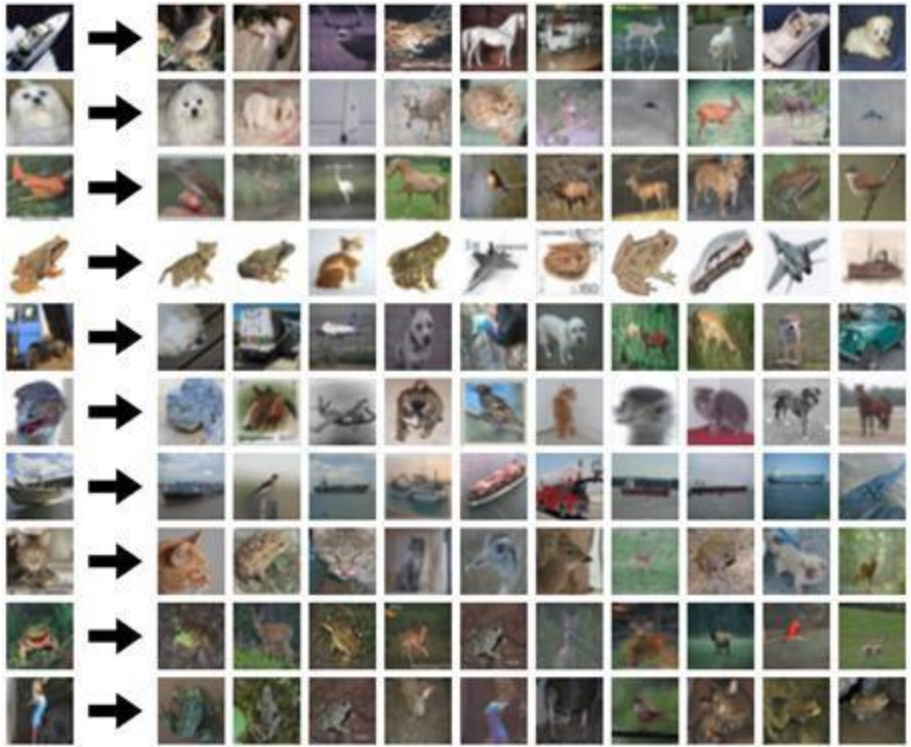
Example of
5-fold cross-validation
for the value of k.

Each point: single
outcome.

The line goes
through the mean, bars
indicated standard
deviation

(Seems that k ~= 7 works best
for this data)

What does this look like?



What does this look like?

k-Nearest Neighbor with pixel distance never used.

- Distance metrics on pixels are not informative

Original

Occluded

Shifted (1 pixel)

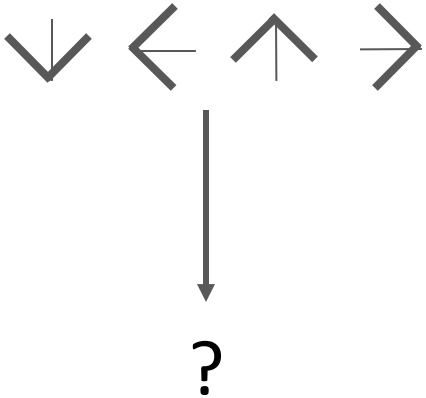
Tinted

(All three images on the right have the same pixel distances to the one on the left)

Attempts have been made

Find edges

Find corners



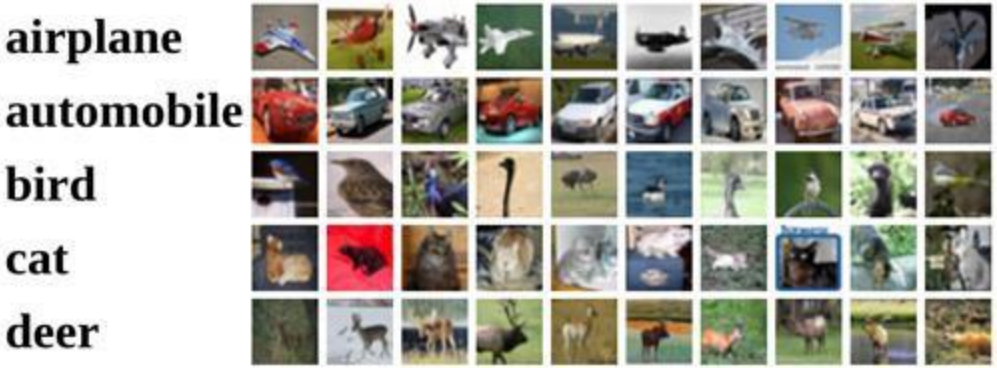
Modern computer vision algorithms

Machine Learning: Data-Driven Approach

1. Collect a dataset of images and labels
2. Use Machine Learning algorithms to train a classifier
3. Evaluate the classifier on new images

```
def train(images, labels):
  # Machine learning!
  return model
```

```
def predict(model, test_images):
  # Use model to predict labels
  return test_labels
```



Parametric Approach

Image

Array of 32x32x3 numbers
(3072 numbers total)

$f(x, W)$

W

parameters
or weights

10 numbers giving
class scores

Parametric Approach: Linear Classifier

$$f(x,W) = W\,x$$

Image

Array of 32x32x3 numbers
(3072 numbers total)

$f(x,W)$

10 numbers giving
class scores

W

parameters
or weights

Parametric Approach: Linear Classifier

$$\boxed{f(x,W)} = \boxed{W}\boxed{x}$$

3072x1

10x1 10x3072

Image

Array of 32x32x3 numbers
(3072 numbers total)

$$f(x,W)$$

10 numbers giving class scores

W

parameters
or weights

Parametric Approach: Linear Classifier

$$\underbrace{f(x,W)}_{10\times 1} = \underbrace{W}_{10\times 3072}\,\overbrace{x}^{3072\times 1} + \underbrace{b}_{10\times 1}$$

Image

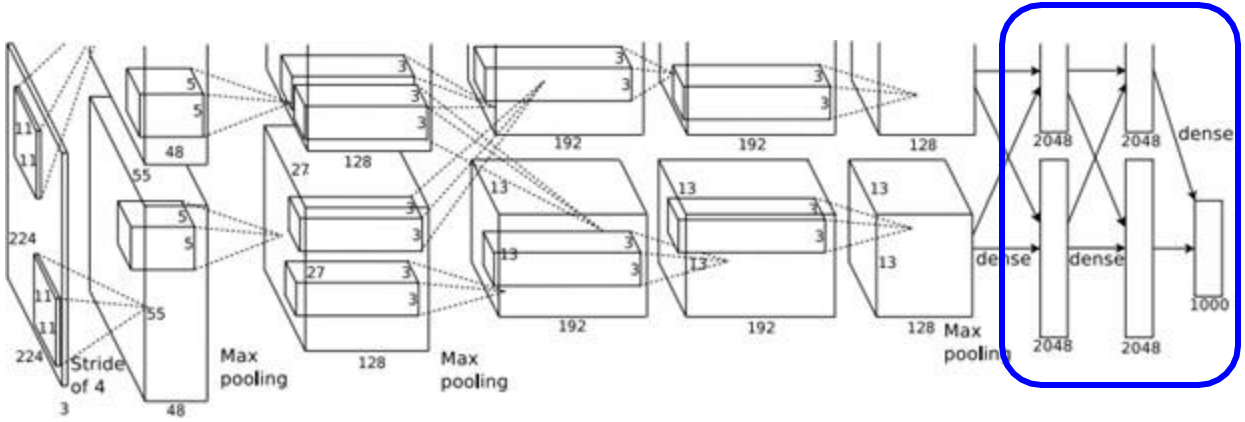
Array of 32x32x3 numbers
(3072 numbers total)

$$\longrightarrow f(x,W) \longrightarrow$$

10 numbers giving class scores

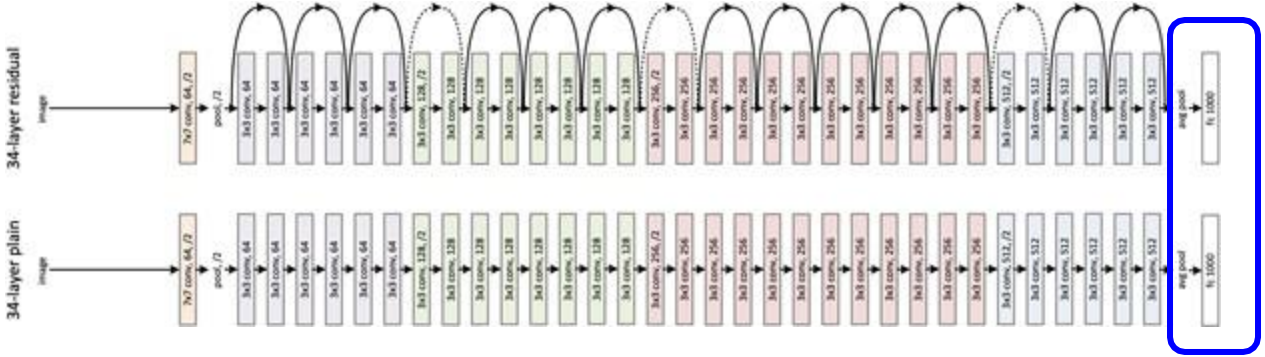
W

parameters
or weights



[Krizhevsky et al. 2012]

Linear layers



[He et al. 2015]

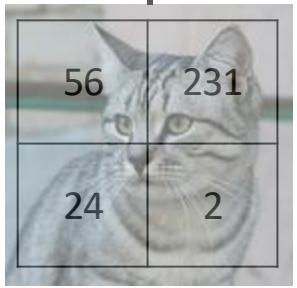
Recall CIFAR10

50,000 training images
each image is 32x32x3

10,000 test images.

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)

Flatten tensors into a vector

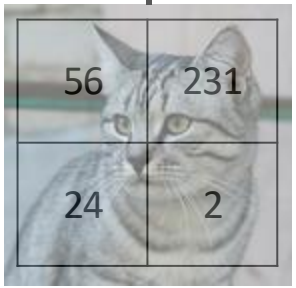


Input image

56
231
24
2

Example with an image with 4 pixels, and 3 classes (cat/dog/ship) <u>Algebraic Viewpoint</u>

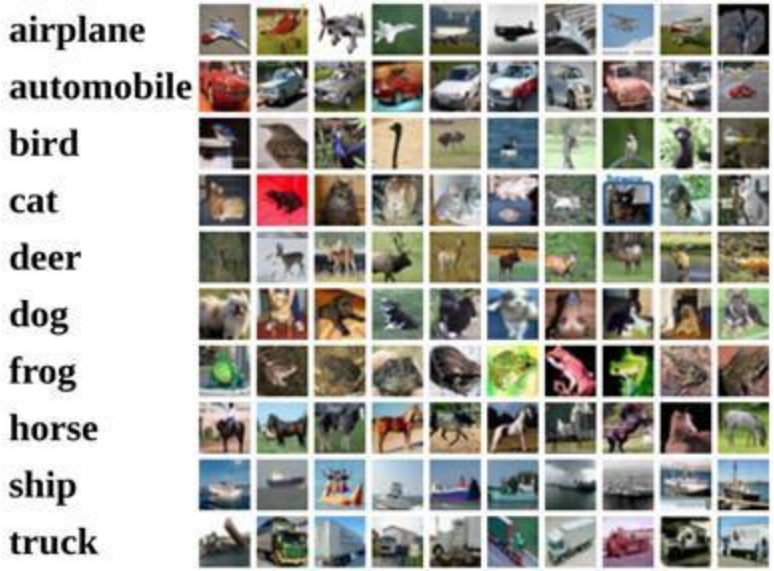
Flatten tensors into a vector

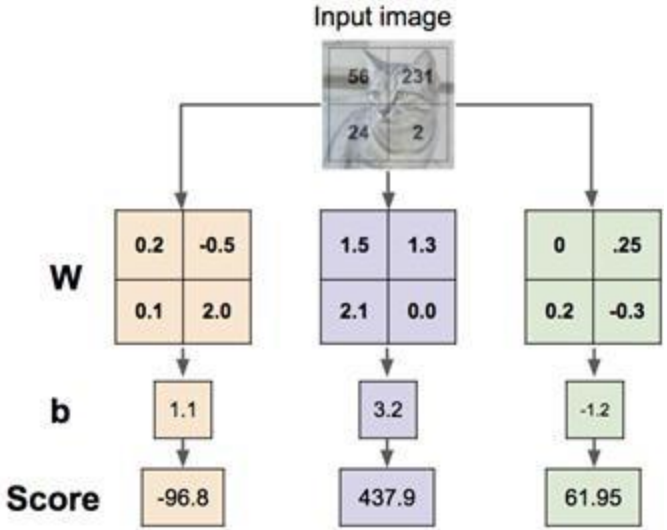


Input image

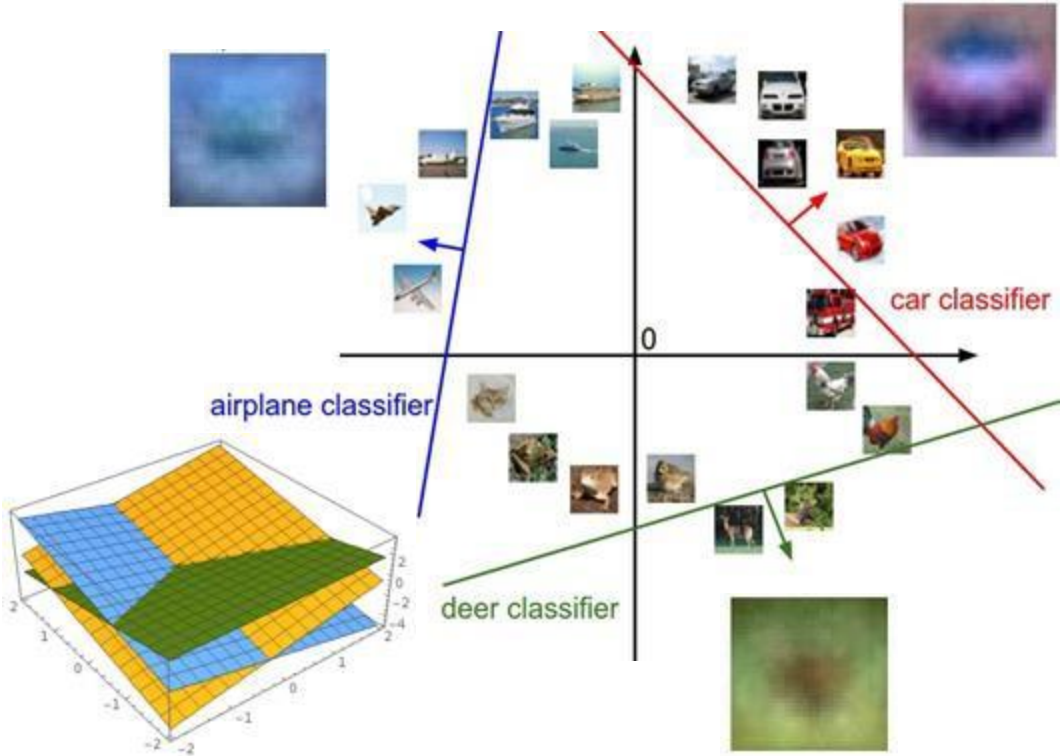
$$\underbrace{\begin{bmatrix} 0.2 & -0.5 & 0.1 & 2.0 \\ 1.5 & 1.3 & 2.1 & 0.0 \\ 0 & 0.25 & 0.2 & -0.3 \end{bmatrix}}_{W} \begin{bmatrix} 56 \\ 231 \\ 24 \\ 2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1.1 \\ 3.2 \\ -1.2 \end{bmatrix}}_{b} = \begin{bmatrix} -96.8 \\ 437.9 \\ 61.95 \end{bmatrix} \begin{matrix} \text{Cat score} \\ \text{Dog score} \\ \text{Ship score} \end{matrix}$$

Interpreting a Linear Classifier





Interpreting a Linear Classifier



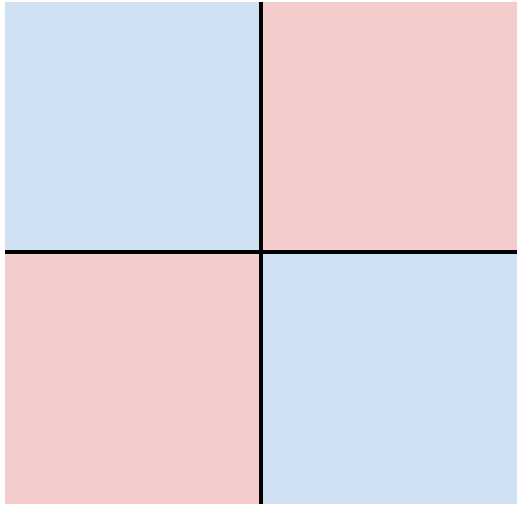
$$f(x,W) = Wx + b$$

Array of 32x32x3 numbers
(3072 numbers total)

Hard cases for a linear classifier

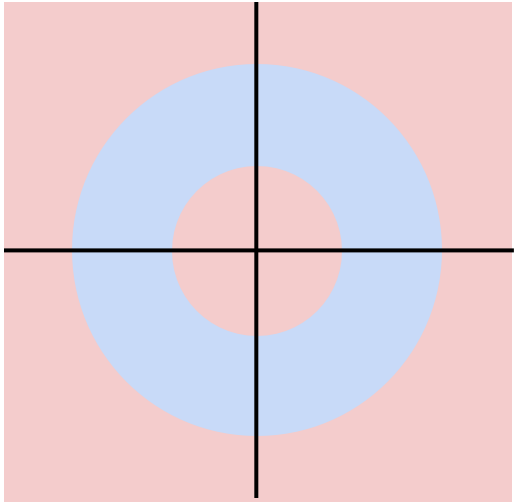
Class 1:
First and third quadrants

Class 2:
Second and fourth quadrants



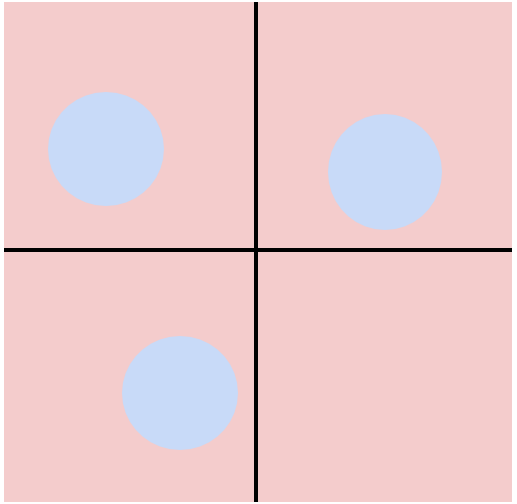
Class 1:
1 <= L2 norm <= 2

Class 2:
Everything else



Class 1: Three modes

Class 2:
Everything else



Linear Classifier – Choose a good W

airplane	-3.45	-0.51	3.42
automobile	-8.87	**6.04**	4.64
bird	0.09	5.31	2.65
cat	**2.9**	-4.22	5.1
deer	4.48	-4.19	2.64
dog	8.02	3.58	5.55
frog	3.78	4.49	**-4.34**
horse	1.06	-4.37	-1.5
ship	-0.36	-2.09	-4.79
truck	-0.72	-2.93	6.14

1. Define a loss function that quantifies our unhappiness with the scores across the training data.
2. Come up with a way of efficiently finding the parameters that minimize the loss function. (optimization)

Example: 3 training examples, 3 classes

cat	3.2	1.3	2.2
car	5.1	4.9	2.5
frog	-1.7	2.0	-3.1

A loss function tells how good our current classifier is

Given a dataset of examples

$$\{(x_i, y_i)\}_{i=1}^{N}$$

Where x_i is image and y_i is (integer) label

Loss over the dataset is an average of loss over examples:

$$L = \frac{1}{N} \sum_i L_i(f(x_i, W), y_i)$$

Softmax Classifier (Multinomial Logistic Regression)

Want to interpret raw classifier scores as probabilities

$$s = f(x_i; W)$$

$$P(Y = k | X = x_i) = \frac{e^{s_k}}{\sum_j e^{s_j}}$$

Softmax Function

Probabilities must be >= 0

Probabilities must sum to 1

$$L_i = -\log P(Y = y_i | X = x_i)$$

	Unnormalized log-probabilities / logits	exp →	unnormalized probabilities	normalize →	probabilities	compare ←	Correct probs
cat	3.2		24.5		0.13		1.00
car	5.1		164.0		0.87		0.00
frog	-1.7		0.18		0.00		0.00

Softmax Classifier (Multinomial Logistic Regression)

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	Unnormalized log-probabilities / logits	exp →	unnormalized probabilities	normalize →	probabilities
cat	3.2		24.5		0.13
car	5.1		164.0		0.87
frog	-1.7		0.18		0.00

→ $L_i = -\log(0.13)$
$= 2.04$

Maximum Likelihood Estimation
Choose weights to maximize the likelihood of the observed data

Softmax Classifier (Multinomial Logistic Regression)

cat	3.2
car	5.1
frog	-1.7

Want to interpret raw classifier scores as probabilities

$$s = f(x_i; W)$$

$$P(Y = k | X = x_i) = \frac{e^{s_k}}{\sum_j e^{s_j}}$$

Softmax Function

Maximize probability of correct class

$$L_i = -\log P(Y = y_i | X = x_i)$$

Putting it all together:

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

Q1: What is the min/max possible softmax loss L_i?

Q2: At initialization all s_j will be approximately equal; what is the softmax loss L_i, assuming C classes?